

DS 413: Introduction to Statistical Learning

Course Code	DS 413
Course Name	Introduction to Statistical Learning
L-T-P-C	3-1-0-4
Prerequisites	DS201 - Data Handling and visualization or equivalent, DS303 – Statistical Foundations of Data Science or equivalent, DS402 – Matrix Computations for Data Science; From 2022 UG Batches
Intended for	Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines
Approval	54th BoA (May 2024)

Course Contents

- **Introduction to learning from data:** Introduction to supervised learning and unsupervised learning. (2 Lectures)
- **Supervised learning: Regression:** Linear regression models and least squares, Shrinkage methods: ridge regression and the LASSO. (6 Lectures)
- **Supervised learning: Classification:** Logistic regression, nearest neighbour's method, Bayes classifier with unimodal and multimodal density - maximum likelihood estimation, expectation-maximization (EM) algorithm; decision trees, support vector machines (SVMs), basics of neural networks. (8 Lectures)
- **Model Assessment and Selection:** Bias, variance and model complexity, The Bayesian approach, AIC and BIC, cross-validation, bootstrap methods, hypothesis testing, confidence intervals, significance testing. (6 Lectures)
- **Unsupervised learning:** Introduction to association rules, clustering, and dimension reduction. (6 Lectures)

Laboratory Modules

Lab to be conducted on a 2-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should be appropriately be sequenced for synchronization with the laboratory.

Text Books

1. Hastie, T., Tibshirani, R. and Friedman, J., **The Elements of Statistical Learning: Data Mining, Inference, and Prediction**, 2nd Edition, 12th Reprint, Springer, 2017.

References

1. Duda, R. O., Hart, P. E. and Stork, D. G., **Pattern Classification**, John Wiley, 2001.
2. Bishop, C. M., **Pattern Recognition and Machine Learning**, Springer, 2006.
3. Theodoridis, S. and Koutroumbas, K., **Pattern Recognition**, Academic Press, 2009.